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Elevated expression of *rsmI* can act as a reporter of aminoglycoside resistance in *Escherichia coli* using kanamycin as signal molecule

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ABSTRACT

We aimed to design and analyse expressional response of endogenous and exogenous 16S rRNA methyl transferase genes under sub inhibitory concentration stress of different clinically relevant aminoglycoside antibiotics in *Escherichia coli* to identify an endogenous marker. One hundred twenty nine aminoglycoside resistant *E. coli* of clinical origin were collected for detection of 16S rRNA methyl transferase genes by PCR assay and each gene type was cloned within *E. coli* JM107. Parent isolates were subjected to plasmid elimination by SDS treatment. Expression analysis of both acquired and endogenous 16S rRNA methyl transferase genes were performed by quantitative real-time PCR in clones and parent isolates under aminoglycoside stress (4 mg/L). Majority of the isolates were harbouring *rmtC* (35/129), followed by *rmtB* (32/129), *rmtA* (21/129), *rmtE* (13/129), *armA* (11/129) *rmtF* (9/129) and *rmtH* (8/129). Plasmid was successfully eliminated for all the isolates with 6% of SDS. Expression analysis indicates that kanamycin, tobramycin and netilmicin stress could increase the expression of 16S rRNA methyltransferese genes. In the presence of kanamycin stress the expression of *rsmI* was consistently elevated for all the wild type isolates and clones tested. Except for isolates harbouring *rmtB* and *rmtC* expression of *rsmE* and *rsmF* was increased in the presence of all aminoglycosides. For all the cured mutants it was apparently observed that expression of endogenous methyl transferases were marginally increased. Elevated expression of constitutive *rsmI* can be used as a potential biomarker for detection of acquired 16S rRNA methyl transferase mediated aminoglycoside resistance by using sub inhibitory concentration of kanamycin as signal molecule.

1. Introduction

Aminoglycosides are broad spectrum bactericidal drugs that bind to A-sites of 16S rRNA and prevent decoding of mRNA which results into loss of translational accuracy (Lioy et al., 2014). One of the prominent mechanisms of resistance to this group of antimicrobials is acquired 16S rRNA methyl transferases (*armA, rmtA, rmtB, rmtC, rmtD, rmtE, rmtF, rmtG, rmtH and npmA*) which methylate G1405 residue in 16S rRNA (Liou et al., 2006). These different types of 16S rRNA methyltransferases are acquired through plasmids in bacteria. However, there also exists endogenous 16S rRNA methyl transferases which post transcriptionally modifies (methylates) smaller sub unit of rRNA and those are designated as *rsmE,rsmF,rsmI* and *rsmH* in *E.coli* (Basturea et al., 2012). This is also known that both exogenous and endogenous 16S rRNA methyl transferases interferes with others activity (Lioy et al., 2014). Therefore, it would be imperative to know how these both types of methyl transferases transcriptionally respond by induced expression when exposed to aminoglycoside antibiotics. The information obtained may be important to identify a reporter for aminoglycoside resistance instead of performing several multiplex PCR assays targeting multiple number of resistance genes. Thus, the present study is designed to investigate expression of endogenous and exogenous 16S rRNA methyl transferase genes under sub inhibitory concentration stress of different clinically relevant aminoglycoside antibiotics.

2. Materials and methods

2.1. Bacterial strains

A total of 129 aminoglycoside resistant *E.coli* isolates harbouring

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different 16S rRNA methyl transferase genes were selected for this study which were isolated from patients who were admitted to different wards or attended outpatient departments of Silchar Medical College and Hospital, Silchar, India from October 2018–September 2019. Aminoglycoside resistance was determined based on MIC testing by agar dilution method, the result of which was interpreted as per CLSI guideline 2017. Further 35 isolates were selected representing acquired 16S rRNA methlytransferase types (5 isolates for each gene types) were used for qRT-PCR based expression study (supplementary table S1).

2.2. Detection of 16S rRNA methyl transferase genes by PCR assay

The presence of different 16S rRNA methyl transferase genes viz.; *rmtA*, *rmtB, rmtC rmtE rmtF,rmtH armA* and *npmA* were determined by PCR assay using the primers (*rmtA*-F 5′CTAGCGTCCATCCTTTCCTC3′) and (*rmtA*-R 5′TTTGCTTCCATGCCCTTGCC3′), (*rmtB*-F5′GGAATTCCATATGAACATCAACGATGCC3′ and *rmtB*-R 5′CCGCTCGAGTCCATTCTTTTTTATCAAGT3′), (*rmtC*-F5′ CGAAGAAGTAACAGCCAAAG3′ and *rmtC*-R 5′GCTAGAGTCAAGCCAGAAAA3′), (*rmtE*-F 5′ATGAATATTGATGAAATGGTTGC3′) and (*rmtE*-R 5′TGATTGATTTCCTCCGTTTTTG 3′),(*rmtF*-F 5′GCGATACAGAAAACCGAAGG 3′) and *rmtF*-R 5′ACCAGTCGGCATAGTGCTTT3′) (*rmtH*-F 5′ AATGACCATTGAACAGGCAGC 3′) and (*rmtH*-R 5′TCAAGCTGGGTTTGGCTGGA3′), (*armA*-F 5′GGTGCGAAAACAGTCGTAGT 3′) and (*armA*-R 5′TCCTCAAATATCCTCTATGT 3′) and (*npmA*-F 5′CGGGATCCAAGCACTTTCATACTGACG3′ and *npmA*-R CGGAATTCCAATTTTGTTCTTATTAGC3′) and reaction conditions were as described earlier(Wu et al., 2009, Davis et al., 2010, Hidalgo et al., 2013, Bueno et al., 2013, O'Hara et al., 2013).The amplified products were further confirmed by sequencing.

2.3. Cloning of acquired 16S rRNA methyl transferase genes

Whole gene including native promoter was amplified from 35 selected isolates (supplementary table S1) harbouring *rmtA, rmtB, rmtC, rmtE, rmtF, rmtH,* and *armA*. The amplified products were further purified by MinElute PCR Purification Kit (Qiagen, Hilden, Germany). The product was ligated in to pGEM-T vector (Promega, Madison, USA), and transferred to *E. coli* JM107 by heat shock method. Transformants were selected based on their growth on screen agar containing gentamicin.

2.4. Plasmid elimination

Plasmid elimination assay was carried out with 35 isolates (supplementary material tableS1) harbouring *rmtA,rmt*B, *rmtC,rmtE,rmtF,rmtH* and *armA* was performed in Luria-Bertani broth. The isolates were treated with sodium dodecyl sulfate (SDS). For SDS treatment 2%,4%,6%,8% and 10% of SDS was added to 5 mL of LB broth, which was then inoculated with 50 μl of a bacterial suspension and kept in a shaker incubator at 37 °C for overnight incubation (Carattoli et al., 2005). After treatment, 30 μl of the bacterial suspension was grown on LB agar with and without kanamycin and elimination of the plasmid was estimated by calculating the number of cells grown on medium with and without aminoglycoside. Kanamycin resistance was used as the selective marker for plasmid curing analysis. PCR was performed further to check whether *armA/rmt* genes were eliminated from the parent types.

2.5. Expression analysis of acquired 16S rRNA methyl transferse genes

Quantitative realtime PCR based expression study of acquired 16S rRNA methyltransferse genes against sub inhibitory concentration of aminoglycoside (amikacin, tobramycin, netilmicin, kanamycin and gentamicin) stress was initiated by inoculating the parental isolates (supplementary table S1) (five organisms of each gene type) and clones (supplementary table S1) harbouring *rmtA,rmtB, rmtC,rmtE, rmtF,rmtH* and *armA* respectively in Luria Bertani broth (Hi-media, Mumbai, India) with and without aminoglycosides at a concentration of 4 mg/L. It was incubated for a period of 16 h and RNA was isolated using Qiagen RNease Mini Kit (Qiagen, Germany), immediately it was reverse transcribed into cDNA by QuantiTect® reverse transcription kit (Qiagen, Germany). The cDNA was quantified by Picodrop (Pico 200, Cambridge, UK) and real time PCR was performed using Power Sybr Green Master Mix (Applied Biosystem, Warrington, UK) in Step One Plus real time detection system (Applied Biosystem, USA) using a set of primers (*rmtA*-RT-F- 5′CTTTGACGATGCCCTAGCGT3′ and *rmtA*-RT-R-5′CAGCTTAGGCGATTTGTGCC3′), (*rmtB*-RT-F 5′CTGCGGTCTTAACCCCTTGG3′ and *rmtB*-RT-R- 5′CCCCCAATCCCTGGTGGATA3′) (*rmtC*-RT-F-5′CGAACACTCGTTGAGAACGC3′ and *rmtC*-RT-R- 5′AAGATCACTCTCGCCTGACG3′), (*rmtE*-RT-F-5′ TTAGGCGACATTGAAGCGGT3′ and *rmtE*-RT-R-5′ GTGCAGTAATGGCAACGCAA3′), (*rmtF*-RT-F-5′ CGTATGTCCCGAGACGGTG3′ and *rmtF*-RT-R-5′TGATCTGGTGCAGATGCGTG3′) (*rmtH*-RT-F-5′ GGCTTAAACCCGCTGATGCT3′ and *rmtH*-RT-R-5′ CCCGTGTTTTTGGGCATAGG3′) and *armA*-RT-F 5′ TGGGGGTCTTACTATTCTGCCT3′ and *armA*-RT-R-5′ TTGCGACTCTTTCATTCGTCG 3′). In the PCR reaction, a housekeeping gene (RpslE) was used as internal control *(rpslE*-F 5'GCAAAAACGTGGCGTGGCGTATGTACTC 3′ and *rpslE*-R 5′ TTCGAAACCGTTAQTCAGACGAA 3′) (Swick et al., 2011). Relative quantification was determined by $2^{-\Delta\Delta Ct}$ method for all the tested genes as per the software provided by the manufacturer where, $\Delta\Delta CT = (C_{T,\ Target} - C_{T,\ Actin})_{Time\ x} - (C_{T,\ Target} - C_{T,\ Actin})_{Time\ 0}$. Time x means any time point and Time 0 is the 1× expression of the target gene. Further, the mean C_T values for both the internal control and target genes were determined at time 0. The mean, standard deviation (SD) and cumulative value (CV) were then calculated from the triplicate sample values at each time point (Iyer et al., 1999). An increase or decrease in expression was interpreted while comparing the RQ value of the corresponding genes under antibiotic exposure with the RQ value of same gene without any antibiotic pressure (RQ value is the mean expression level of the gene in qPCR).

2.6. Expression analysis of the endogenous methyl transferase genes

Expression analysis of the 5 endogenous methyl transferase genes namely *rsmE,rsmF,rsmH* and *rsmI* were determined by quantitative real time PCR assay for wild type isolates (five representatives of each acquired 16 s methyl transferases), clones and the cured mutants without any antibiotic pressure (*ΔrmtA ΔrmtB, ΔrmtC,ΔrmtD, ΔrmtE, ΔrmtH* and *ΔarmA)*. Additionally, expressional response of rsm genes against aminoglycoside exposure (4 mg/L) was also performed for wild types and clones. The total RNA was isolated using QIAGEN Rneasy Mini Kit (QIAGEN, Germany) following manufacturer's instructions. Estimation of the isolated RNA was performed using Picodrop (Pico200, Cambridge, UK) and further cDNA was synthesized using prepared mRNA as template via Qiagen Reverse Transcription Kit (QIAGEN, Germany). Quantitative Real-time PCR of the cDNA prepared was done using power Sybrgreen PCR master mix reagents kit (Applied Biosystems, Austin, USA) in StepOnePlus quantitative Real Time-PCR (Applied Biosystems, USA) in triplicates using specific primers such as *rsmE*(F) 5′CCATGACTGCCCGCTATCAA3′;*rsmE* (R) 5′AGCGCAGTTGTCTCTGTACG3′,*rsmF*(F) 5′CCGATTTTCTCGGTTGGGGA3′; *rsmF*(R) 5′AGGCAAGAGCAATAACCGCT3′, *rsmH* (F) 5′ CTCACGTCTGATCCTCTCGC3′; *rsmH* (R) 5′ CAATAGTCTTCGCAACGGCG3′ and *rsmI* (F) 5′ AATCCCGCTACGTGGTTCTG3′v; *rsmI* (R) 5′ TCTTCCTTTACCCACGCCAG3′. The relative expression of the methyl transferase genes in each interval with and without antibiotic pressure was determined by $2^{-\Delta\Delta Ct}$ method for all the tested genes. *E. coli* ATCC 25922 was used as a reference strain which was grown for 4 h without any type of antibiotic pressure. An increase or decrease in expression was interpreted while comparing the RQ value of the corresponding genes under antibiotic exposure with the RQ value of same gene without any antibiotic pressure.

2.7. Susceptibility assessment of parent strain and cured mutants

Susceptibility testing towards aminoglycosides was done for both parent and cured mutants. Kirby Bauer disc diffusion and minimum inhibitory concentration determination by broth dilution was performed and results were interpreted as per CLSI guideline 2017.

2.8. Statistical analysis

The changes in expression of both acquired and endogenous methyltransferase genes in response to different aminoglycoside stresses at 4 μg/ml concentration was analysed using one-way ANOVA and Tukey-kramer (Tukey's W) multiple comparison test. Differences were considered statistically significant at when p value is ≤ 0.05.

3. Results

3.1. Molecular characterization of 16S rRNA methyl transferase genes and their cloning and plasmid elimination

In the present study we have observed seven different types of acquired 16S rRNA methyl transferase genes among isolates and analysed each type to investigate further the endogenous methyl transferases. Among 129 aminoglycoside resistant isolates, twenty one were harbouring *rmtA*, thirty two *rmtB*, thirty five *rmtC*, thirteen *rmtE*, nine *rmtF*, eight *rmtH* and eleven isolates were found to carry *armA*. For all the selected isolates (supplementary table S1) plasmids harbouring rmt and armA gene types were eliminated and was found that it reverted into a susceptible phenotype (Table 1). While performing qRT-PCR assay, it was observed that kanamycin, tobramycin and netilmicin stress was able to increase expression of *rmtA* (Fig. 1 and Fig. S1) whereas, in case of *rmtB*, kanamycin stress was able to elevate the expression (Fig. 2 and Fig. S2). It was also observed that expression of *rmtC* was increased in the presence of kanamycin and tobramycin (Fig. 3 and Fig. S3). However, in case of gene types *rmtE, rmtF* and *rmtH*, aminoglycoside stress was unable to increase the expressional patterns (Figs. 4–6 and Figs. S4–S7), whereas, tobramycin stress could increase expression of *armA* (Fig. 7 and Fig. S7).

3.2. Expression analysis of the endogenous methyl transferase genes

While analysing expression of endogenous methyl transferases under sub inhibitory aminoglycoside stress, it was observed that in the presence of aminoglycoside stress, especially in the presence of kanamycin, expression of *rsmI* was consistently elevated for all the isolates (Figs. 1–7 and Fig. S1 to S7). Besides this, expression of *rsmH* was also increased in the presence of kanamycin, gentamicin and netilmicin in the isolates harbouring *rmtA,rmtB* and *rmtC* (Figs. 1–3). Expression of *rsmE* and *rsmF* was increased in the presence of all aminoglycosides but not in all the isolates (except isolates harbouring *rmtB* and *rmtC*). A noticeable observation was that expression of endogenous methyl transferases increased marginally for all cured mutants. ANOVA test showed expression of endogenous *rsmE, rsmF, rsmI* and *rsmH* in response to *armA, rmtB, rmtC* and *rmtA* with aminoglycoside stress were statistically significant (p- value of *armA* = 0.04, *rmtB* = 0.007, *rmtC* = 0.007 and *rmtA* = 0.04). However, ANOVA test showed that p value of *rmtE* = 0.5, *rmtF* = 0.3, *rmtH* = 0.5 in response to endogenous *rsmE, rsmF, rsmI* and

Table 1
Susceptibility profile of parent and cured mutants towards aminoglycosides (interpreted as per CLSI guidelines 2017)

Isolate ID	Resistance genes	Wild type /Zone of inhibition(diameter) in mm/MIC value in mg/L										Cured mutants/Zone of inhibition(diameter) in mm/ MIC value in mg/L									
		amikacin		gentamicin		netilmicin		tobramycin		kanamycin		amikacin		gentamicin		netilmicin		tobramycin		kanamycin	
		D	M	D	M	D	M	D	M	D	M	D	M	D	M	D	M	D	M	D	M
JB2	*armA*	12	64	10	32	11	32	5	32	5	64	21	8	17	2	16	4	16	4	20	8
JB11	*armA*	11	64	9	32	11	32	5	32	5	64	20	8	17	4	17	4	16	4	20	4
JB32	*armA*	13	64	9	16	11	32	7	16	5	64	18	8	19	2	17	4	16	4	19	4
JB19	*armA*	12	128	8	16	10	32	7	16	5	64	25	8	20	2	17	4	17	4	19	8
JB7	*armA*	10	64	10	16	10	32	0	16	7	64	25	8	20	4	17	4	19	4	19	4
JB16	*rmtA*	12	64	9	32	9	32	0	16	6	64	19	8	20	2	20	4	19	4	25	8
JB39	*rmtA*	12	64	10	16	9	32	6	32	6	64	19	8	21	4	20	4	20	4	23	8
JB78	*rmtA*	11	64	10	32	10	32	6	16	6	64	20	8	25	4	20	4	20	4	25	4
JB21	*rmtA*	10	64	9	16	11	32	6	16	0	64	20	8	17	2	17	2	20	4	19	4
JB10	*rmtA*	11	64	9	16	10	32	3	16	0	64	25	4	17	2	17	4	17	2	20	4
JB57	*rmtB*	10	128	11	16	10	32	4	16	0	64	25	8	20	2	17	4	17	4	19	4
JB42	*rmtB*	12	64	10	16	9	32	3	16	7	6	21	8	20	2	21	4	17	4	25	4
JB51	*rmtB*	12	64	10	32	9	32	5	32	0	64	21	4	17	2	25	2	20	4	19	8
JB34	*rmtB*	13	128	11	32	9	32	5	32	0	64	21	4	25	4	23	2	19	4	19	8
JB72	*rmtB*	11	128	11	16	8	32	6	32	9	64	27	4	21	4	23	4	17	2	20	8
JB65	*rmtC*	11	128	10	32	9	32	6	32	5	64	27	8	20	4	23	4	16	2	25	8
JB3	*rmtC*	13	64	10	16	9	32	8	16	5	64	27	8	17	2	17	2	16	2	19	8
JB8	*rmtC*	12	64	9	16	8	32	8	16	5	64	25	4	17	2	17	2	16	4	18	8
JB93	*rmtC*	10	64	9	32	9	32	8	16	3	64	25	4	17	2	17	2	17	2	20	4
JB95	*rmtC*	9	64	9	32	8	32	0	16	3	64	25	8	19	4	20	4	16	4	25	4
JB23	*rmtE*	11	64	8	32	8	32	0	16	3	64	20	8	21	4	20	2	16	4	18	4
JB37	*rmtE*	12	64	5	32	8	32	0	32	0	64	20	8	25	2	25	4	16	4	19	4
JB14	*rmtE*	12	128	10	32	10	32	0	32	3	64	25	4	21	2	20	4	17	4	25	8
JB25	*rmtE*	9	128	11	16	9	32	6	16	0	64	26	4	20	2	20	2	20	4	19	4
JB29	*rmtE*	10	64	10	16	9	32	7	16	0	64	27	4	19	4	18	4	18	2	18	8
JB36	*rmtF*	10	64	11	16	9	32	5	16	0	6	20	4	17	4	18	2	19	4	20	4
JB72	*rmtF*	11	64	9	16	11	32	0	32	5	64	21	4	17	4	19	4	20	4	20	8
JB49	*rmtF*	12	64	9	32	10	32	6	16	3	64	20	8	17	4	17	4	16	4	25	4
JB56	*rmtF*	12	64	10	32	11	32	6	16	5	64	20	8	18	4	24	4	17	2	19	4
JB93	*rmtF*	11	128	9	32	9	32	0	32	5	64	22	8	18	2	24	2	17	4	18	4
JB121	*rmtH*	12	128	10	32	9	32	10	32	5	64	25	4	25	2	23	2	17	4	25	4
JB115	*rmtH*	10	128	9	16	9	32	0	16	5	64	25	4	25	2	20	4	17	4	19	4
JB6	*rmtH*	11	64	8	16	11	32	0	16	8	64	27	4	25	4	17	2	20	2	20	8
JB15	*rmtH*	11	128	11	16	10	32	5	16	8	64	20	8	25	4	18	4	16	2	25	8
JB29	*rmtH*	12	64	10	16	10	32	5	32	5	64	25	4	21	2	20	2	16	4	19	4

D = Zone diameter in mm for disc diffusion method.
M = Minimum inhibitory concentration value in mg/L.

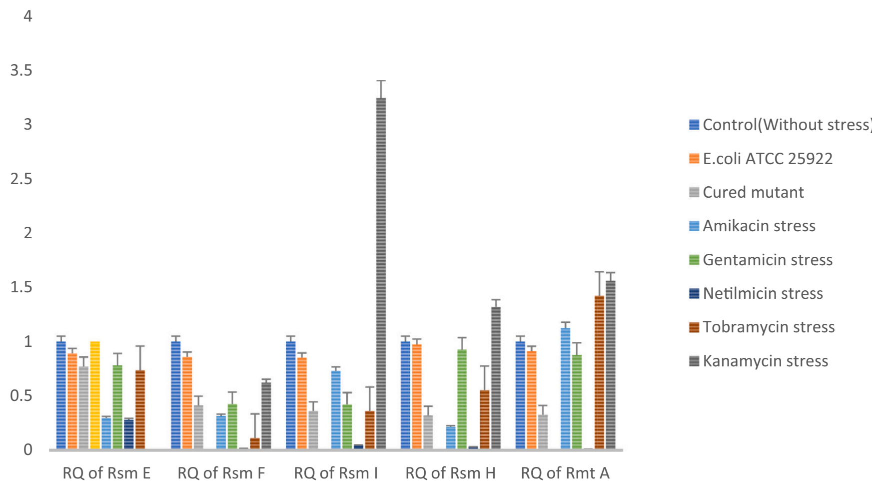

Fig. 1. Expression of *rmtA,rsmE,rsmF,rsmI* and *rsmH* gene with and without stress condition in *E.coli* harbouring *rmtA*.

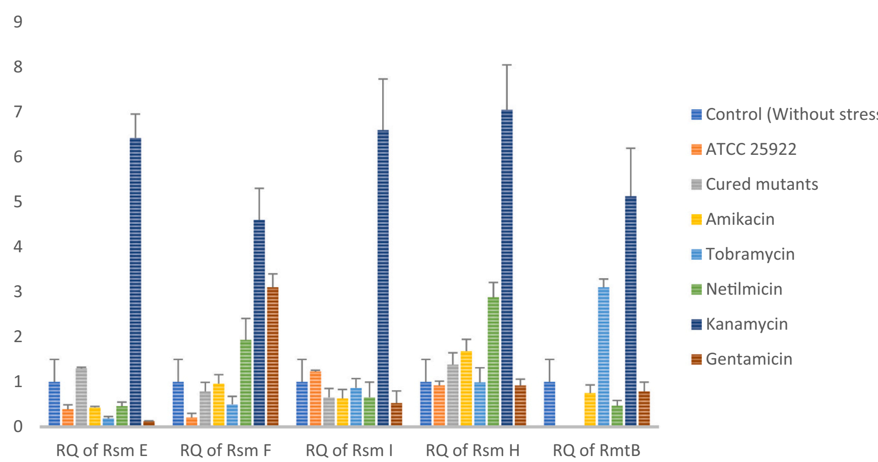

Fig. 2. Expression of *rmtB,rsmE,rsmF,rsmI* and *rsmH* gene with and without stress condition in *E.coli* harbouring *rmtB*.

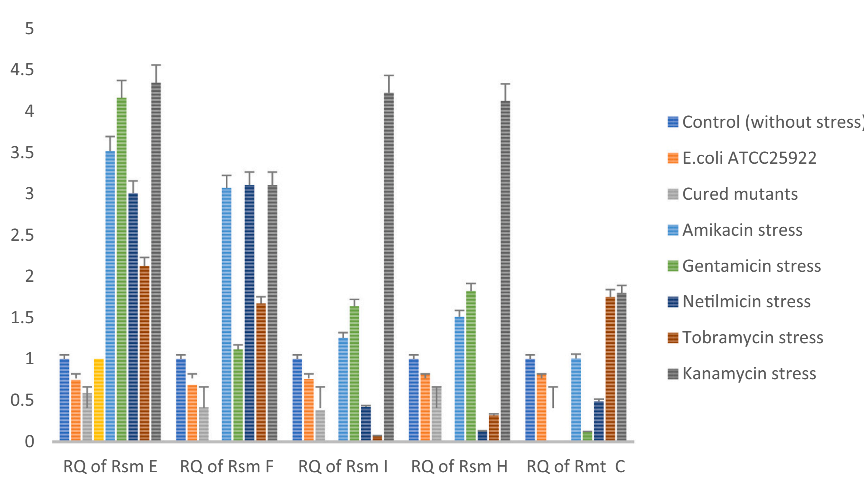

Fig. 3. Expression of *rmtC,rsmE,rsmF,rsmI* and *rsmH* gene with and without stress condition in *E.coli* harbouring *rmtC*.

rsmH with aminoglycoside stress.

4. Discussion

Mechanism of aminoglycoside resistance is diverse and many AMEs are present which cannot be detected by a single multiplex PCR. Several sets of PCR assays have to be developed to identify the resistance gene. Thus, designing a PCR assay which will report aminoglycoside resistance irrespective of the types of AMEs will be a key to rapid detection expressional pattern for *rmt* genes and *armA* under aminoglycoside

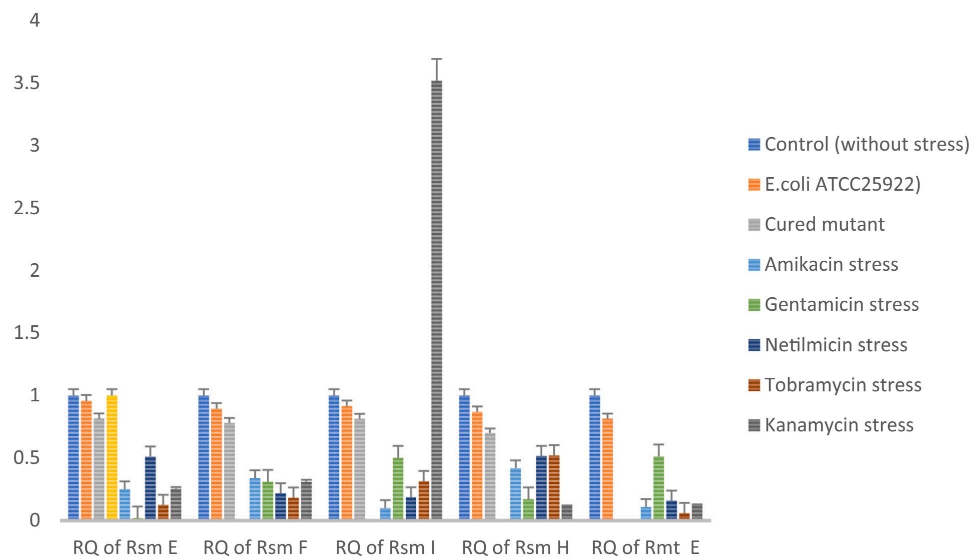

Fig. 4. Expression of *rmtE,rsmE,rsmF,rsmI* and *rsmH* gene with and without stress condition in *E.coli* harbouring *rmtE*.

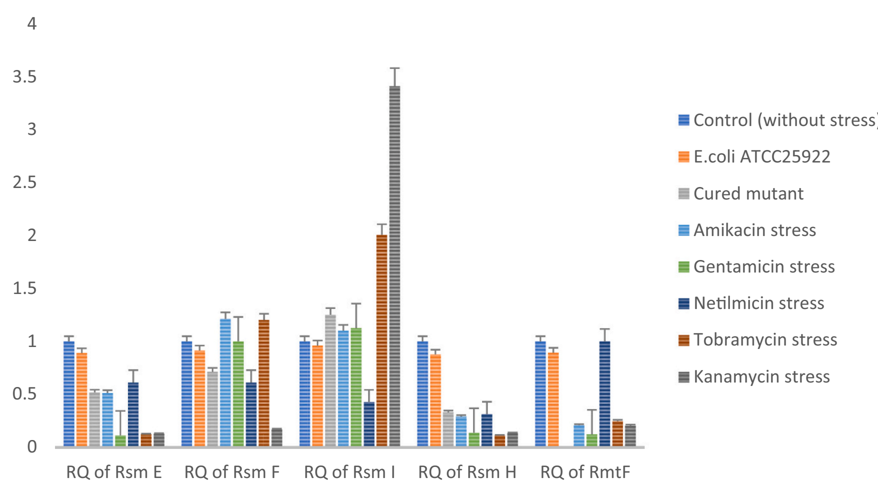

Fig. 5. Expression of *rmtF,rsmE,rsmF,rsmI* and *rsmH* gene with and without stress condition in *E.coli* harbouring *rmtF*.

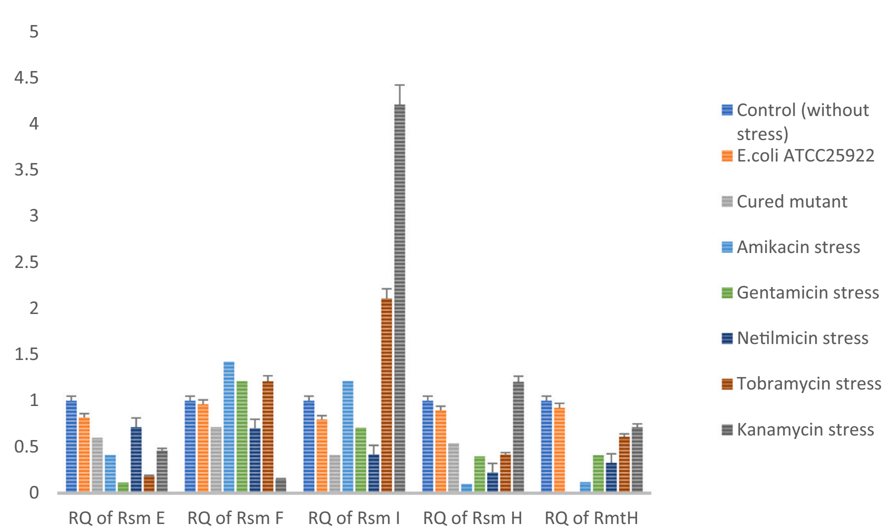

Fig. 6. Expression of *rmtH,rsmE,rsmF,rsmI* and *rsmH* gene with and without stress condition in *E.coli* harbouring *rmtH*.

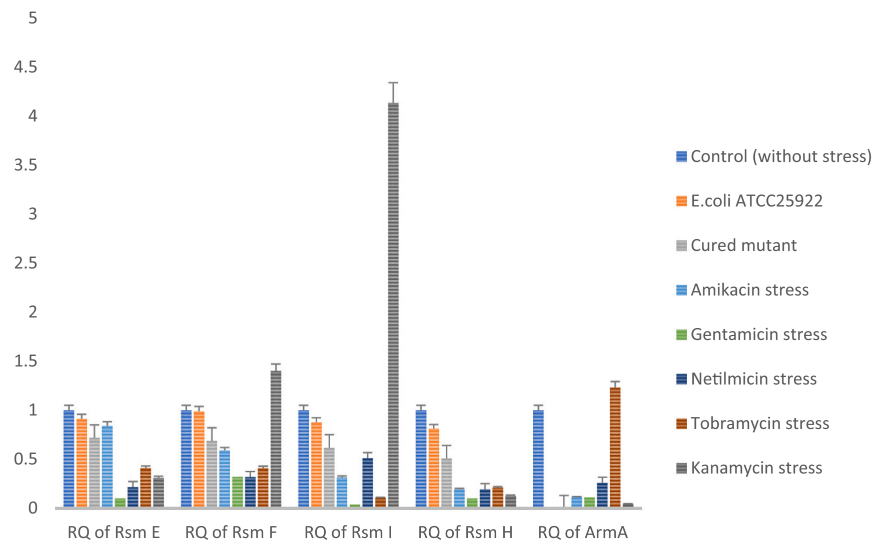

Fig. 7. Expression of *armA, rsmE, rsmF, rsmI* and *rsmH* gene with and without stress condition in *E. coli* harbouring *armA*.

stress was inconsistent. However, in case of endogenous methyl transferases *rsmI* showed consistent upregulation when kanamycin stress was administered. So far, no other study is there in this aspect to compare. However, this study was able to compare the pattern of expression of both exogenous and endogenous 16S rRNA methyl transferases under aminoglycoside stress. Kanamycin, a potent aminoglycoside binds in the helix 44 of prokaryotic 16S rRNA and interact with residues G1491 and A1408 by forming hydrogen bond (Salian et al., 2012). *ArmA* and *rmt* methyl transferases methylates G1405 residue in the 16S rRNA (Gutierrez et al., 2012), whereas endogenous methyl transferase *rsmI* methylates C1402 residue of 16S rRNA (Kimura and Suzuki, 2010). In a study, it was observed that *armA*/rmt blocks activity of *rsmI* at C1402 leaving it non methylated (Virginia et al., 2014). Additionally it was also shown that non methylation at C1402 delay in cell growth and altering ribosome functioning (Kimura and Suzuki, 2010). This is probably a reason why sub inhibitory concentration of kanamycin induces expression of *rsmI* in order to enhance methylation process under antibiotic exposure. The interplay between housekeeping and acquired 16S rRNA methyltransferase genes against sub-inhibitory concentration of kanamycin may be under regulation of the promoter system of plasmid mediated resistance system. It was observed in previous study that the promoter of *rmtC* was originated from IS*Ecp1* elements (Gutierrez et al., 2012). In the current study, we too incorporated native promoters of respective rmt genes in preparation of clones for expression studies However, further investigation is required to confirm this claim. This may also be hypothesized that in normal condition ie., in antibiotic free environment *armA*/rmt impedes methylation at C1402 by *rsmI*, which although impair cell growth does not have any larger effect on cell whereas, when exposed to aminoglycoside (kanamycin), the methylation at C1402 is initiated as observed by elevated expression of *rsmI* in this study. The susceptibility testing of cured mutants showed that elimination of plasmid carrying acquired 16S rRNA methyl transferase reverted the isolates in to aminoglycoside susceptible phenotypes (Table 1).

5. Conclusion

The present study also underscored that endogenous 16S rRNA methyl transferases does not have any direct role in aminoglycoside resistance as all cured mutants were reverted back to a susceptible phenotype. The study shows that elevated expression of constitutive *rsmI* can be used as a potential biomarker for detection of acquired 16S rRNA methyl transferase mediated aminoglycoside resistance by using sub inhibitory concentration of kanamycin as signal molecule.

Author contributions

JW performed the experimental work, data collection and analysis and prepared the manuscript. AB Supervised the research work and participated in designing the study and drafting the manuscript. DP have done data analysis. MS participated in sample collection and part of experiments. DD participated in experiment designing and manuscript correction. All authors read and approved the final manuscript.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.meegid.2022.105229.

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